



UNIVERSITY OF ABOMEY-CALAVI

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LABORATORY OF BIOMATHEMATICS AND FOREST ESTIMATIONS (LABEF)



SPECIES DISTRIBUTION MODELING OF GREVY'S ZEBRA (Equus grevyi) IN KENYA (2055)

EXAM

Submitted by
OTIENO Brian

Supervisor

Prof. Dr Ir Adandé Belarmain Fandohan
Full Professor of Forestry

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LIST OF ABBREVIATIONS AND ACRONYMS

Table 1: Description of Bioclimatic and Environmental Variables

Variable	Description
Bio 1	Annual Mean Temperature
Bio 2	Annual Mean Diurnal Range (Mean of monthly (max temp - min temp))
Bio 3	Isothermality (Bio2 / Bio7)
Bio 4	Temperature Seasonality (Standard Deviation)
Bio 4a	Temperature Seasonality (Coefficient of Variation)
Bio 5	Max Temperature of Warmest Month
Bio 6	Min Temperature of Coldest Month
Bio 7	Annual Temperature Range (Bio5 - Bio6)
Bio 8	Mean Temperature of Wettest Quarter
Bio 9	Mean Temperature of Driest Quarter
Bio 10	Mean Temperature of Warmest Quarter
Bio 11	Mean Temperature of Coldest Quarter
Bio 12	Annual Precipitation
Bio 13	Precipitation of Wettest Month
Bio 14	Precipitation of Driest Month
Bio 15	Precipitation Seasonality (Coefficient of Variation)
Bio 16	Precipitation of Wettest Quarter
Bio 17	Precipitation of Driest Quarter
Bio 18	Precipitation of Warmest Quarter
Bio 19	Precipitation of Coldest Quarter
dm	Distance to Major Roads or Settlements (proxy for human disturbance)
llds	Livestock Density or Landcover/Land Degradation Score
pet	Potential Evapotranspiration (climate-driven water loss potential)
mimq	Moisture Index of Moist Quarter
mi	General Moisture Index
miaq	Moisture Index of Arid Quarter

Chapter 1

INTRODUCTION

Species distribution modeling is an indispensable tool in modern ecology and conservation biology, offering a robust framework for understanding the spatial arrangement of species and forecasting their responses to environmental change. These models operate on the fundamental assumption that a species' presence in a specific location is contingent upon the availability of suitable environmental conditions that can support its survival, foraging, and reproduction (Sohl, 2014). Ecological niche models utilize species occurrence data and current environmental factors to pinpoint the environmental conditions that significantly limit a species' range, estimating a species' current environmental situation using environmental covariate data in a correlative statistical modeling framework Koot et al., 2022. These models, by correlating species occurrences with environmental variables, allow for the extrapolation of potential habitats, thereby providing insights into the ecological requirements and limitations of the species Soberón & Peterson 2005.

The application of species distribution modeling has become increasingly prevalent in biogeographical research, addressing questions related to range dynamics, drivers of diversity patterns, and the delineation of ecoregions Franklin, 2023. The precision of species distribution models in forecasting climatic niches, especially for endangered species, is vital for conservation efforts Khadka & James, 2017. Species distribution models extrapolate from associations between point occurrences and environmental data sets to identify areas of predicted presence Soberón & Peterson, 2005. By integrating species occurrence data with a suite of environmental predictors, such as climate, topography, and land cover, these models can generate spatially explicit predictions of habitat suitability.

Kenya has a tropical climate with hot and humid conditions along the coast, and cooler temperatures in the highlands. Inland, temperatures are cooler, ranging from 16 °C (60 °F) to 24 °C (75 °F). The country experiences two rainy seasons, the long rains from March to May and the short rains from October to December. The hottest months are January and February, with temperatures reaching up to 30 °C (86 °F). The coldest months are July and August, with temperatures dropping to as low as 10 °C (50 °F).



Figure 1.1: *Map of Kenya*

Source: Encyclopaedia Britannica, 2011

Grevy's zebra, known as *Equus grevyi* is the largest and most endangered of the three zebra species, distinguished by its narrow stripes, large ears, and white belly (IUCN, 2020). Endemic to the arid and semi-arid regions of northern Kenya and parts of Ethiopia, this species has experienced a dramatic population decline due to habitat loss, competition with livestock, poaching, and climate change Williams et al., 2021 therefore applying species distribution modeling is very important to this species. Once widespread across the Horn of Africa, Grevy's zebra now occupies less than 3% of its historical range, with fewer than 2,500 individuals remaining in the wild Grevy's Zebra Trust, 2023.



Figure 1.2: *Equus grevyi*

Source: Grevy's Zebra Trust, 2023

Species Distribution Modeling (SDM) provides a powerful approach to predict a species' potential geographic range based on occurrence data and environmental variables. This project uses SDM to identify current and future suitable habitats for Grevy's zebra, incorporating climate projections from the AFRICLIM dataset under different Representative Concentration Pathways (RCPs).

1.1 Objectives

1. To develop a Species Distribution Model for the current distribution of *Equus grevyi* .
2. To identify critical habitats and potential range expansions for *Equus grevyi* using species distribution modeling.
3. To assess the impact of climate change on the future distribution of *Equus grevyi* by incorporating climate scenario projections into SDMs.

Chapter 2

METHODS

2.1 Study Area

Kenya lies in the East of the African continent and South of the Horn of Africa. It expands over a size of approximately $584000 km^2$(Orodho, 2006). According to the WRI (2007), almost 40% of Kenya's ecosystems are savannah and grassland. More than $75,238 km^2$ (or 13%) of the land surface is set aside as national parks and game reserves (WDPA, 2008) The study focuses on northern Kenya, such as Samburu, Isiolo, and Marsabit counties, which host the largest remaining populations of Grevy's zebra (Equus grevyi) (Grevy's Zebra Trust, 2023). This region lies within the arid and semi-arid lands (ASALs) of Kenya, characterized by highly variable rainfall $(300–600 mm/year)$, seasonal river systems (Ewaso Ng'iro, Tana River), and a mix of shrublands and acacia grasslands (KWS, 2022). The area is under increasing pressure from climate variability, livestock grazing, and habitat fragmentation, making it a critical zone for conservation prioritization. Our study area covered 680,657 observations of 21 covariates in Kenya administrative units.

2.2 Data Sources

Georeferenced occurrence records of *Equus grevyi* were obtained from the Global Biodiversity Information Facility (GBIF)[GBIF.org (2025). GBIF Occurrence Download. https://doi.org/10.15468/dl.e6pqsa]. The 19 bioclimatic variables both present and future (seasonal summary variables relating to temperature (tbio) and moisture (mbio)) at $1-km$ resolution were sourced from AFRICLIM v3.0 Platts et al. (2015) [https://www.york.ac.uk/environment-geography/research/kite/resources/]. The Administrative areas (boundaries) of Kenya data was obtained DIVA-GIS by [https://diva-gis.org/data.html].

Software and tools that were used include R (packages: raster, dismo, rjava, sf, geodata, terra) and MaxEnt software.

Table 1 shows different environmental covariates downloaded and their meanings.

The DIVA-GIS program was used to extract the 19 climate variables data of Kenya, while Maxent described the present E. grevyi distribution using the location data as presence training points. The climate envelopes were then mapped, and changes in the distribution range were analyzed for present and future climates. The climate variables in the same grid as presence points were extracted, averaged, and the difference sought from the current to 2055. The robustness of the model developed was validated using cross tabulation, one of the methods available in the Maxent software.

Chapter 3

RESULTS AND DISCUSSION

The total population of *Equus grevyi* was found to be 717, which was subjected to data cleaning, and 37 duplicates and 1 missing data were found; therefore, the total occurrences of *Equus grevyi* considered in this study are 682.

Below is a graph of *Equus grevyi* distribution in Kenya 2025

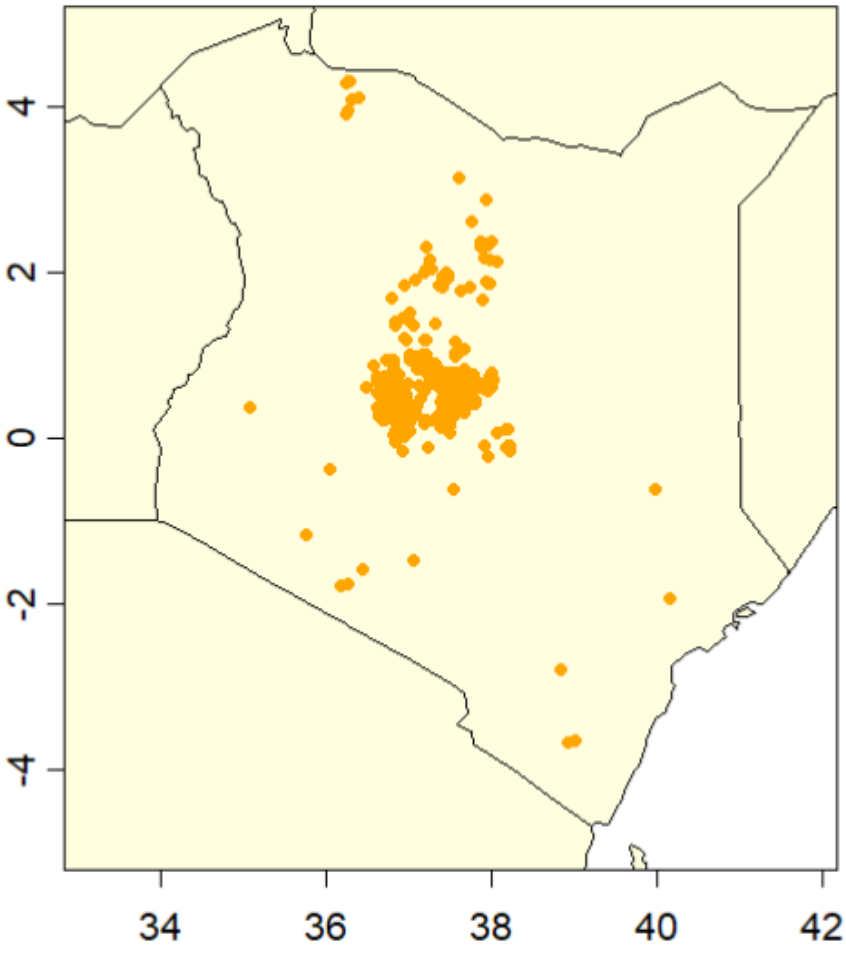


Figure 3.1: The distribution of *Equus grevyi* in Kenya 2025

3.1 Model Selection

In this study We considered three main models with different environmental variables affecting *Equus grevyi* in order to establish a suitable habitat. The first model was built with 9 variables which were considered after reading the various literature and selecting some of the factors affecting suitabiliy of the habitat. Below are the results we found:

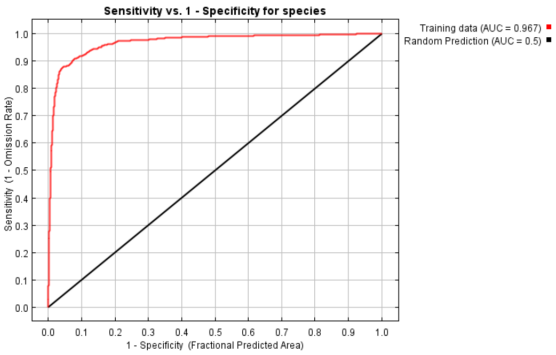


Figure 3.2: The receiver operating characteristic (ROC) curve for E. grevyi

Variable	Percent contribution	Permutation importance
bio4	32.3	25.9
llds	24.8	1.7
bio10	19.9	34.8
bio12	9.8	8.8
pet	6	2.9
mimq	3.3	16.2
bio5	2.1	7.8
bio3	1.6	1.4
bio1	0.2	0.5

Figure 3.3: Contribution of covariates for E. grevyi

The second model was built with all 21 covariates, and we obtained the following results:

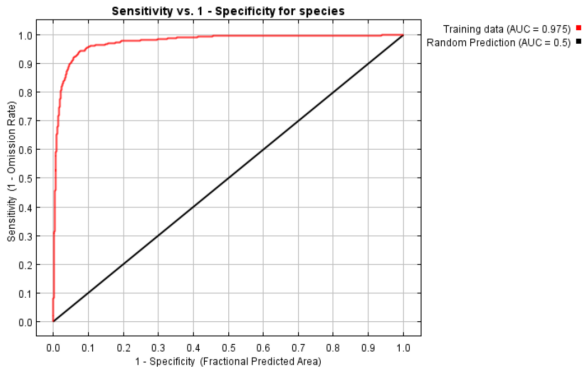


Figure 3.4: The receiver operating characteristic (ROC) curve for E. grevyi (21 variables)

Variable	Percent contribution	Permutation importance
bio2	39.1	3.5
bio4	23	46.5
dm	10.6	10.1
llds	8.8	0.8
bio17	4.5	5.9
bio10	4.2	0.5
bio15	2.7	17.2
bio3	2.3	1.9
pet	1.5	1.3
mimq	1.2	5.2
bio16	0.6	0.6
bio14	0.3	1.3
bio6	0.3	0.1
mi	0.2	0.4
bio7	0.2	1
bio12	0.2	0.8
bio5	0.1	0.9
bio13	0.1	1.5
bio11	0.1	0.4
miaq	0	0
bio1	0	0

Figure 3.5: Contribution of covariates (21) for E. grevyi

The third model we considered was the five covariates which had best perfoemance i.e., (Permutation importance ≥ 5) from the general model (all covariates) table 3.14

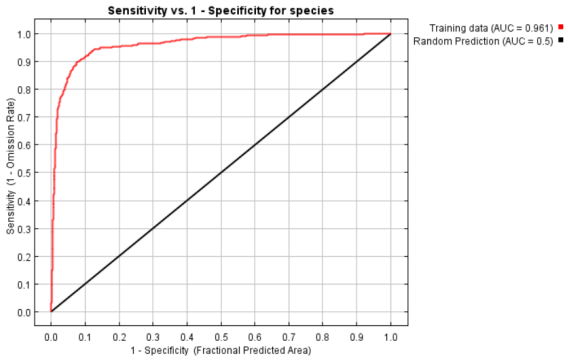


Figure 3.6: The receiver operating characteristic (ROC) curve for E. grevyi (21 variables)

Variable	Percent contribution	Permutation importance
bio4	45.5	51
dm	24	3
mimq	15.9	8.9
bio17	8.1	6.5
bio15	6.4	30.7

Figure 3.7: Contribution of covariates (21) for E. grevyi

In order to select the best model, we decided to compare the three models using the table below

No.	Name of the Model	AUC
1	Model with 9 covariates	0.967
2	Model with 21 covariates	0.975
3	Model with 5 covariates	0.961

Table 3.1: Model Performance Comparison by Area Under Curve (AUC)

From the table 3.1, we conclude that the model with the 21 covariates is the best due to its high level of accuracy (AUC = 0.975). Therefore this model was suitable for prediction of the future habitat of *E. grevyi* in Kenya in the next 30 years (2055)

3.2 Environmental covariates

We obtained the environmental covariate graph for Africa, which was later used to obtain the environmental covariates for kenya as shown below, both current and future (2055), suitable for *Equus grevyi* habitat.

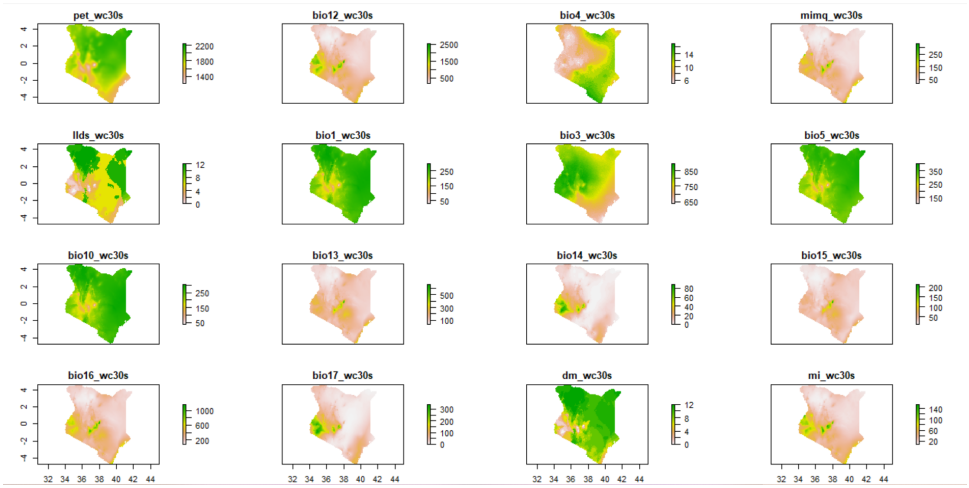


Figure 3.8: Current environmental covariates Kenya (2025)

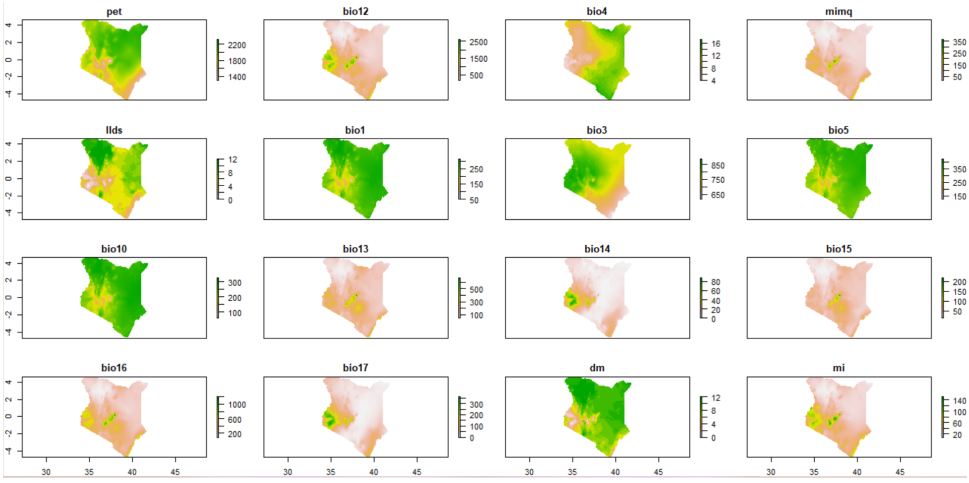


Figure 3.9: Current environmental covariates Kenya (2055)

3.3 Analysis of maxent graphs

All 21 climatic variables were used to train the model. Figure 3.10 is the omission rate and predicted area as a function of the cumulative threshold, which is calculated on The training presence records, on the test records. The closer the Omission on the training samples line is to the predicted omission, the more accurate the generated model

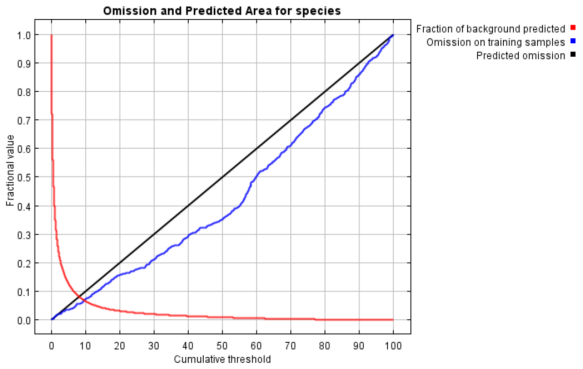


Figure 3.10: The omission rate and predicted area as a function of the cumulative threshold for E. grevyi

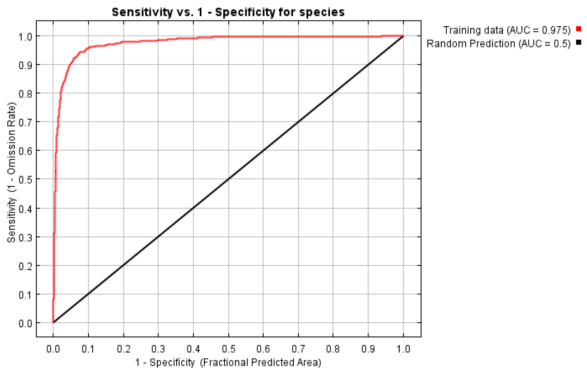


Figure 3.11: The receiver operating characteristic (ROC) curve for E. grevyi (21 variables)

Area Under Curve (AUC) of the Receiver Operating Characteristic (ROC) curve figure 3.11, is a parameter used to evaluate the predictive ability of the generated model. It measures the likelihood that a randomly selected presence point is located in a raster cell with a higher probability value for species occurrence than a randomly selected absence point. The generated model's AUC for training data was 0.975 with a random prediction AUC of 0.5. This is an excellent model as the AUC is more than 0.90 as recommended by (Araújo et al. (2005)). (Philips et al. (2005)) compared Maxent and Genetic Algorithm for Rule-Set Prediction methods in species geographic distribution work. Their results showed that both models were significantly better than random when tested for omission and ROC analysis. They further concluded that Maxtent showed better discrimination of suitable and unsuitable areas of the species in the analysis of AUC.

3.4 Response curve

These curves in figure 3.12 show how each environmental variable affects the Maxent prediction. The curves show how the predicted probability of presence changes as each environmental variable is varied, keeping all other environmental variables at their average sample value.The curves show the marginal effect of changing exactly one variable, whereas the model may take advantage of sets of variables changing together.

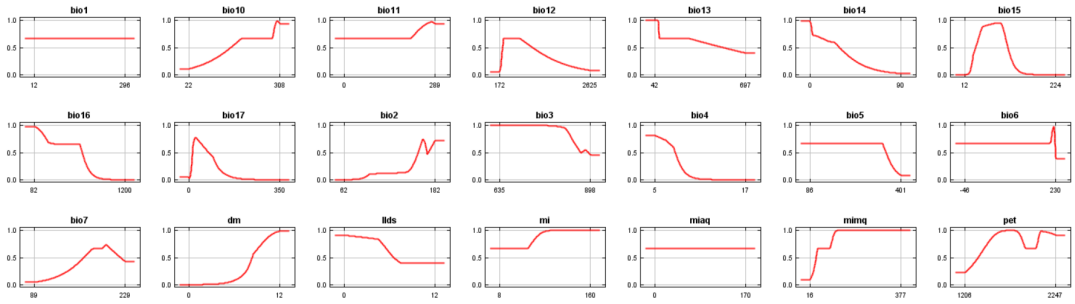


Figure 3.12: Effects of each environmental variable on the Maxent prediction for E. grevyi

Each of the following curves in figure 3.13 represents a different model, namely, a Maxent model created using only the corresponding variable, unlike the above marginal response curves in figure 3.12 . These plots reflect the dependence of predicted suitability both on the selected variable and on dependencies induced by correlations between the selected variable and other variables.

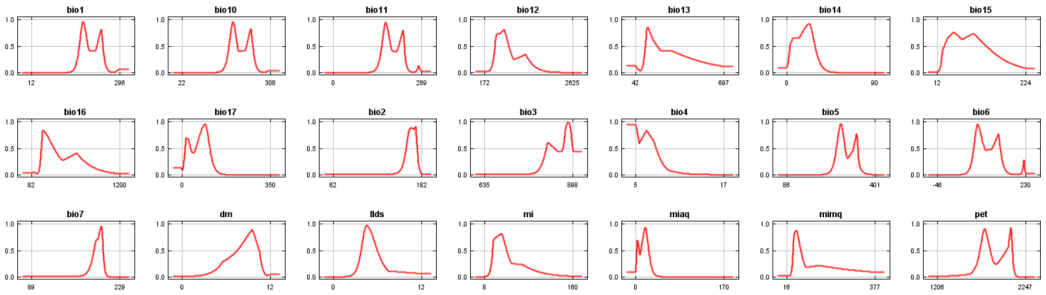


Figure 3.13: Maxent model created using only the corresponding variable for E. grevyi

3.5 Analysis of variable contributions

The model is reevaluated on the permuted data, and the resulting drop in training AUC is shown in the table, normalized to percentages.

Variable	Percent contribution	Permutation importance
bio2	39.1	3.5
bio4	23	46.5
dm	10.6	10.1
llds	8.8	0.8
bio17	4.5	5.9
bio10	4.2	0.5
bio15	2.7	17.2
bio3	2.3	1.9
pet	1.5	1.3
mimq	1.2	5.2
bio16	0.6	0.6
bio14	0.3	1.3
bio6	0.3	0.1
mi	0.2	0.4
bio7	0.2	1
bio12	0.2	0.8
bio5	0.1	0.9
bio13	0.1	1.5
bio11	0.1	0.4
miaq	0	0
bio1	0	0

Figure 3.14: Contribution of covariates for E. grevyi

The percent contribution reflects how much each variable contributed to the final MaxEnt model based on the path taken during model training. It depends on the particular path of the optimization process, while Permutation importance reflects the decrease in model performance (AUC) when the values of a variable are randomly permuted. This measures how much useful information the variable provides, independent of the model path.
From the figure results, temperature seasonality (bio4) has the second highest percent contribution (23%), indicating Grevy's zebra is highly sensitive to seasonal temperature variations and the highest permutation importance (46.5%), meaning it has some unique predictive power.The highest permutation importance (46.5%), suggesting this variable is critical and contains non-redundant information i.e., It's a key driver of habitat suitability.

The bio2 (Mean Diurnal Temperature Range) and llds (landscape variable) contribute 39.1% (highest) but only 3.5% permutation importance and 8.8% with 0.8% permutation importance respectively. This implies that thevariables may be correlated with other variables or spatially autocorrelated. Despite a low training contribution, Moisture index moist quarter (mimq) it still provides unique information not captured by other variables therfore important to retain in the model.

The MaxEnt model reveals that climate factors, particularly temperature-related variables (bio4, bio2, bio15, bio17), dominate Grevy's zebra habitat suitability, collectively contributing 73.1̃% of the model's predictive power. Temperature seasonality (bio4) and warm-season temperatures (bio15 and bio2) are especially critical, while precipitation (bio17) plays a secondary role (5.9%), indicating the species is more sensitive to thermal conditions than rainfall. Non-

climate variables (pet, mimq) have minor but notable effects, suggesting localized or indirect influences on distribution. These findings underscore the importance of temperature stability in conservation planning for this species, which is potentially vulnerable to climate change.

3.6 Analysis of influential response curve

3.6.1 bio4-Temperature Seasonality Curve

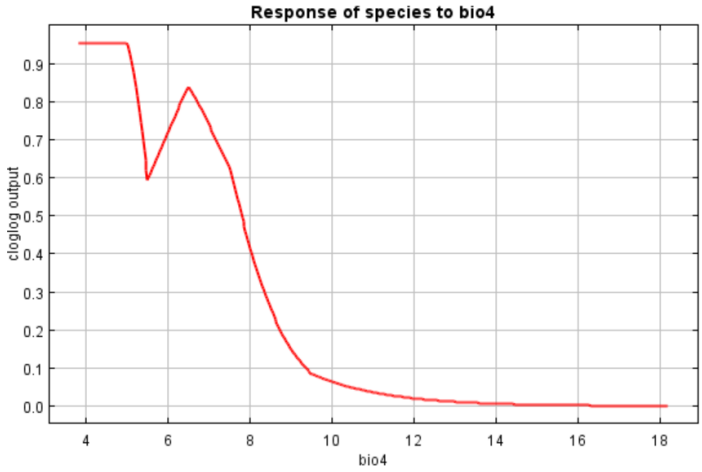


Figure 3.15: bio4-Temperature Seasonality Curve

Ecological Implication:
The response curve for Bio4 (Temperature Seasonality) reveals that Grevy's zebra exhibits a strong preference for habitats with low annual temperature variability. Habitat suitability is highest when temperature seasonality values range between approximately 4 and 6, indicating that stable thermal conditions are optimal for the species. Beyond this range, particularly above a value of 7, habitat suitability declines sharply, reaching near-zero levels as temperature fluctuations increase. This pattern suggests that Grevy's zebra is highly sensitive to seasonal temperature extremes, likely due to the impact of such variability on vegetation productivity, water availability, and overall habitat stability. Consequently, conservation planning should prioritize areas with minimal temperature seasonality to support the long-term persistence of Grevy's zebra populations under changing climate conditions

3.6.2 Bio15 (Precipitation Seasonality) response curve

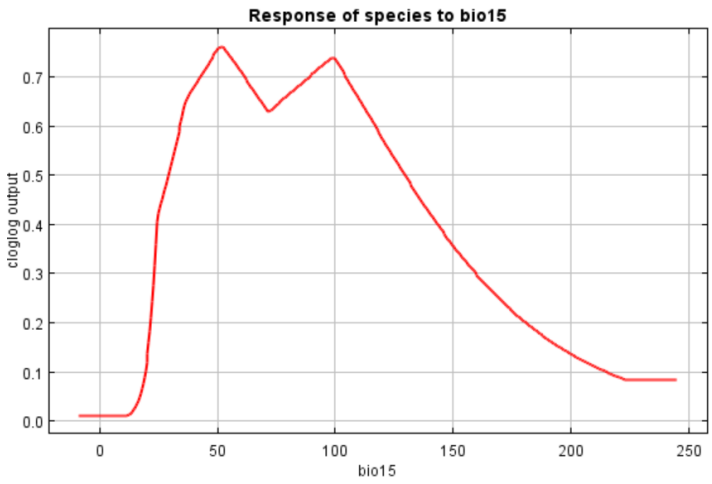


Figure 3.16: Bio15 (Precipitation Seasonality) response curve

Ecological Implication:
The response curve for Bio15 (Precipitation Seasonality) shows that Grevy's zebra exhibits the highest habitat suitability in areas with moderate precipitation variability. Suitability increases sharply from near zero when Bio15 values rise from 0 to around 50–60, peaking at approximately 55 and again near 90. These peaks suggest that Grevy's zebra favors regions where rainfall patterns are somewhat seasonal but not extreme. Beyond this range, especially after a Bio15 value of 100, habitat suitability declines steadily, reaching minimal levels as precipitation variability becomes more pronounced.

3.6.3 Bio2 (Mean Diurnal Range) response curve

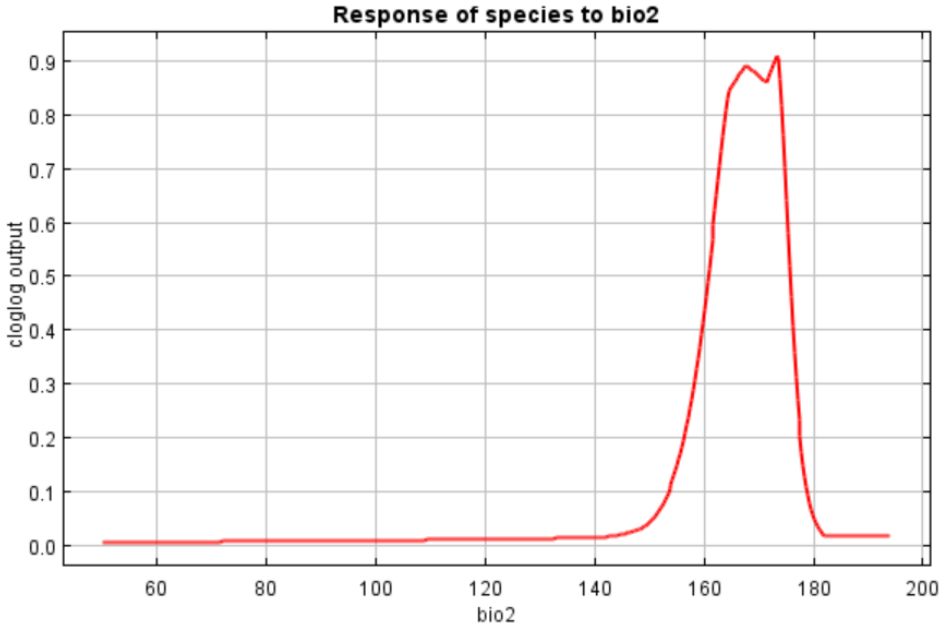
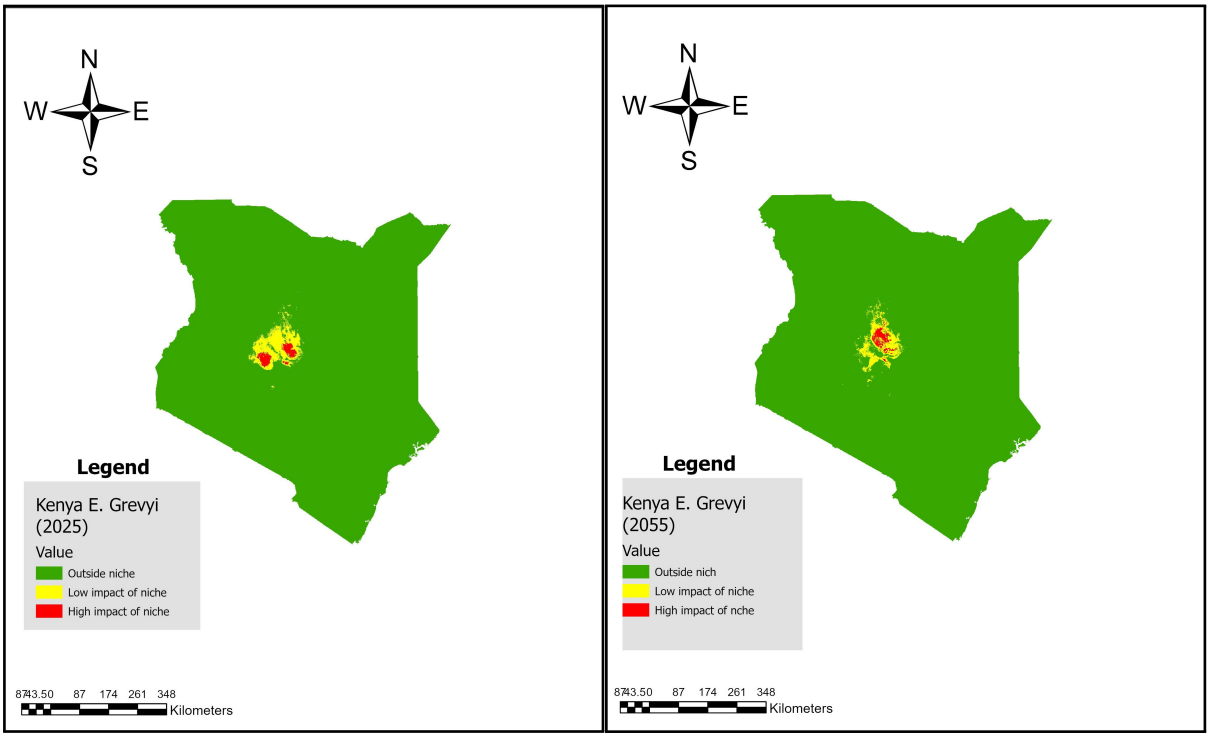


Figure 3.17: Bio2 (Mean Diurnal Range) response curve

Ecological Implication:
The response curve for Bio2 (Mean Diurnal Range) reveals that Grevy's zebra is highly sensitive to daily temperature fluctuations, with habitat suitability peaking sharply at values between approximately 165 and 170. Suitability is negligible outside this narrow range, indicating a strong preference for environments with moderate diurnal thermal variation. This pattern suggests that extreme day–night temperature differences may negatively impact the species, emphasizing the importance of thermally stable habitats for its conservation.

3.7 Prediction of current and future (2055) habitat of *Equus grevy*



(a) Suitable habitat for E. Grevyi (2020)

(b) Future habitat prediction (2055)

Chapter 4

CONCLUSION

The MaxEnt model indicates that habitat suitability for Grevy's zebra in Kenya is primarily determined by temperature-related factors, particularly temperature seasonality (bio4) and the mean temperature of the warmest quarter (bio2 and bio15), which together provide the most critical and non-redundant information for predicting suitable environments. These findings suggest that Grevy's zebra favors habitats with specific and stable thermal conditions, especially during warmer periods. While some landscape variables like llds show high model contribution, their low permutation importance implies limited direct influence on habitat suitability. In contrast, variables like the moisture index of the moist quarter (mimq), despite low training contribution, offer unique ecological insights and should be retained. Overall, the results highlight the importance of thermal stability in habitat formation for Grevy's zebra, emphasizing the need for conservation strategies that account for climate sensitivity and the potential impacts of future warming.

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